



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME
DSA0115 Object Oriented Programming with C++
for System Level Programming

OPERATOR OVERLOADING

QUESTIONS 19 / 19 completed

(-)

Write a C++ program to overload the - operator to subtract two complex numbers.

PROGRAM EDITOR

C++ Run Save

```
4 int real, imag;
5 public:
6     Complex(int r = 0, int i = 0) {
7         real = r;
8         imag = i;}
9     Complex operator-(Complex c) {
10        Complex temp;
11        temp.real = real - c.real;
12        temp.imag = imag - c.imag;
13        return temp; }
14    void display() {
15        cout << real << " + " << imag << "i"; }
16 int main() {
17     int r1, i1, r2, i2;
18     cin >> r1 >> i1;
19     cin >> r2 >> i2;
20     Complex c1(r1, i1);
21     Complex c2(r2, i2);
22     Complex c3 = c1 - c2;
23     c3.display();
24     return 0;}
```

OUTPUT

```
3 + 1i
```

INPUT

```
5 4
2 3
```

QUESTIONS 19 / 19 completed

Plus Equal

PLUS EQUAL

Write a C++ program to overload the += operator to add two objects of a user-defined class.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Number {
4     int x;
5 public:
6     void input() {
7         cin >> x; }
8     void operator+=(Number n) {
9         x += n.x; }
10    void display() {
11        cout << x;
12    };
13 int main() {
14     Number a, b;
15     a.input();
16     b.input();
17     a += b;
18     a.display();
19     return 0;
20 }
```

OUTPUT

30

INPUT

10
20

QUESTIONS 19 / 19 completed

Add

ADD

Write a C++ program to overload a function to add two integer numbers and two floating-point numbers separately.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Add {
4 public:
5     int add(int a, int b) {
6         return a + b; }
7     float add(float a, float b) {
8         return a + b;
9     };
10 int main() {
11     Add obj;
12     int a, b;
13     float x, y;
14     cin >> a >> b;
15     cin >> x >> y;
16     cout << obj.add(a, b) << endl;
17     cout << obj.add(x, y);
18     return 0;
19 }
```

OUTPUT

30
50.5

INPUT

10 20
24.6 25.9

QUESTIONS 19 / 19 completed

Maximum

MAXIMUM

Write a C++ program to overload a function to find the maximum value from two integer numbers, two floating-point numbers, and two characters separately.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Maximum {
4 public:
5     int max(int a, int b) {
6         return (a > b) ? a : b;
7     }
8     float max(float a, float b) {
9         return (a > b) ? a : b;
10    };
11 int main() {
12     Maximum obj;
13     int a, b;
14     float x, y;
15     cin >> a >> b;
16     cin >> x >> y;
17     cout << obj.max(a, b) << endl;
18     cout << obj.max(x, y);
19     return 0;
20 }
```

OUTPUT

```
20
5.5
```

INPUT

```
10 20
5.5 3.2
```

QUESTIONS 19 / 19 completed

Concatenate ▾

CONCATENATE

Write a C++ program to overload a function to concatenate two strings and two character arrays separately.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Concatenate {
5 public:
6     string concat(string a, string b) {
7         return a + b; }
8     string concat(string a, string b, string c) {
9         return a + b + c;
10    };
11 int main() {
12     Concatenate obj;
13     string a, b, c;
14     cin >> a >> b;
15     cin >> c;
16     cout << obj.concat(a, b) << endl;
17     cout << obj.concat(a, b, c);
18     return 0;
19 }
```

OUTPUT

```
HelloWorld
HelloWorldC++
```

INPUT

```
Hello World
C++
```